A unified construction of continuously tunable multiqubit elegant joint measurements

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Continuous tuning of the common Bloch radius of an elegant joint measurement requires its local regular-tetrahedral geometry to remain compatible with a complete orthonormal measurement basis. This paper presents an explicit continuous family of elegant joint measurements for every integer $n \geq 5$, using a unified phase construction for odd and even n. The family passes through a specified reference measurement, preserves orthonormality and fixed local tetrahedral directions, and continuously covers the radius interval $[0, 2r_E(n)]$, where $r_E(n) = \sqrt{3}/2^{n-1}$ is the reference radius. Within a fixed phase representation, a complete classification is obtained of the radius laws permitted by straight phase paths with two-valued real generators that pass through the reference measurement and preserve regular tetrahedral geometry with a common radius in its neighborhood. Consequently, the bound $2r_E(n)$ is attained and optimal within this model for every $n \geq 5$, and $n = 5$ is the smallest system size allowing a radius increase in this model. For every even $n \geq 8$, a two-dimensional measurement family is further constructed whose two coordinate sections recover the new radius-increasing family and the known cosine family column by column. The low-dimensional results further reveal the role of the phase path's shape: at $n = 4$ in the same phase representation, even with arbitrary real phase generators, no straight phase path through the reference measurement that preserves the stated geometric conditions in its neighborhood can increase the radius, whereas an explicit curved phase path reaches $(1 + 1/\sqrt{2})r_E(4)$.

I. INTRODUCTION

An entangled projective measurement is constrained both by the orthogonality of its global basis states and by the geometry of their local reductions. The elegant joint measurement (EJM) brings these two features together: the local Bloch vectors occupy the vertices of a regular tetrahedron, while the corresponding global states form an orthonormal basis. Its four local directions coincide with the geometry of a qubit symmetric informationally complete (SIC) measurement [1]. Introduced in the study of quantum networks [2, 3], the EJM at $n = 2$ belongs to a continuous family connecting it to a Bell-state measurement [4]. Members of this family yield violations of bilocal Bell inequalities [4] and full-network-nonlocality criteria [5], and have been implemented in photonic entanglement-swapping experiments [6].

The same two-qubit EJM also arises from maximizing the equal pairwise distinguishability of single-qubit reductions, an approach that admits higher-dimensional bipartite constructions from SIC measurements [7]. More broadly, iso-entangled two-qubit bases have been classified [8], and the extension of a pure state to a complete basis through local unitaries has been studied [9]. Classification by the shared entanglement needed for local implementation provides a further operational perspective [10]. For the present problem, varying the tetrahedral radius changes the single-qubit reduced spectrum and hence the entanglement across each single-qubit-versus-rest cut. Extending this control to more qubits requires the local geometries of all measurement outcomes to remain compatible with global orthogonality.

Pauli-orbit constructions provide a useful setting for this problem [11, 12]. In the associated phase normal form, every diagonal unitary generates a complete orthonormal basis; the remaining constraints concern the local Bloch vectors. Reference [13] gives an explicit reference EJM for every $n \geq 2$, with common radius

$$r_E(n) = \frac{\sqrt{3}}{2^{n-1}}, \tag{1}$$

and an even-qubit phase family with radius $r_E(n)\cos t$ on $|t| \leq \pi/2$. This construction already identifies interference involving a single computational-basis phase correction as the source of the local signal. The odd-qubit analysis in that work leaves an explicit tunable family through the specified reference point open for $n \geq 5$, and identifies a locally isolated reference point at $n = 3$ from a numerical Jacobian rank. Other symmetric multiqubit constructions permit broader local geometries [14, 15]; here the target remains one fixed regular tetrahedron at each qubit, with a tunable common radius across qubits and outcomes.

Here we construct an explicit continuous family with radius $r_E(n)(1 - \sin t)$ for every integer $n \geq 5$. A single Boolean kernel, depending on four binary features, applies to both odd and even numbers of qubits. The family recovers the reference measurement column by column at $t = 0$ and continuously connects zero local polarization to twice the reference radius. This radius increase raises a natural question about its limit. We answer it for all two-valued real phase generators through the reference measurement by classifying the admissible radius laws. The doubled radius is optimal in this model, which includes generators beyond the four-feature construction.

The unified construction also makes it possible to compare the new radius-increasing deformation with the known even-qubit cosine deformation within one phase representation. We examine whether both can be varied

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independently while preserving the local tetrahedra. For every even $n \geq 8$, the answer is an explicit two-parameter family of complete measurements containing the two deformations as exact coordinate sections. At $n = 6$, mixed interference obstructs this extension; in the model that groups inputs by the four joint values of two Boolean functions and assigns one phase to each group, the local solutions consist of two crossing branches.

The lower-dimensional cases clarify why a straight phase path can cease to be sufficient. No straight phase path with a two-valued generator increases the reference radius for $n = 2, 3, 4$. At $n = 3$, we strengthen earlier numerical evidence [13] with an analytic proof of local rigidity. At $n = 4$, even allowing arbitrary real generators prevents every admissible straight line from increasing the radius, but an explicit curved path increases the radius to $(1 + 1/\sqrt{2})r_E(4)$.

Section II develops the phase framework that reduces these questions to local interference conditions. Sections III and IV give the unified construction and its sharp Boolean-model bound. Section V establishes its relation to the even-qubit family, and Sec. VI explains the low-dimensional obstructions and the curved construction at $n = 4$. Proofs of the key steps not developed in the main text are collected in the appendixes.

II. PHASE FRAMEWORK AND LOCAL GEOMETRY

This section develops a phase framework that preserves measurement orthonormality and reduces the local geometry of all columns to the signed components of a seed state. The tunable parameters enter through a diagonal phase matrix. For $n \geq 2$, we use the computational basis $|z_1 \cdots z_n\rangle$, with $z_j \in \mathbb{F}_2$, and the circuit convention of Ref. [13]. Products act from right to left. Write

$$\begin{aligned} C_n &= S(n)H_n, \qquad M[D] = C_n D H^{\otimes n}, \\ S(n) &= \mathrm{CNOT}_{2\to1} \cdots \mathrm{CNOT}_{n\to n-1}, \end{aligned} \tag{2}$$

where H_n acts only on the last qubit and D is diagonal with unit-modulus entries. By unitarity, the columns $|\psi_k\rangle = M[D]|k\rangle$ form a complete orthonormal measurement for every such D. Their local geometry is determined by the seed $|\psi_0\rangle$: writing $k = (e_1, \ldots, e_{n-1}, \sigma)$ with $e \in \mathbb{F}_2^{n-1}$ and $\sigma \in \mathbb{F}_2$, the columns obey the Pauli orbit identity

$$|\psi_{e,\sigma}\rangle = \left[\prod_{h=1}^{n-1} (Z_h Z_{h+1})^{e_h}\right] (X^{\otimes n})^{\sigma} |\psi_0\rangle. \tag{3}$$

This orbit propagates the seed's local Bloch vectors to all columns. To obtain regular tetrahedra with a common radius, it suffices to make the seed's signed local components equal. Our local coordinates are fixed throughout: only the final-qubit Y component is sign-adjusted, $\widetilde{Y}_n = -Y_n$; every other $\widetilde{P}$ equals the corresponding physical Pauli operator. A common signed component $m \geq 0$ therefore means seed Bloch vectors $m(1, 1, 1)$ for $j < n$ and $m(1, -1, 1)$ for $j = n$.

We deform the reference phase

$$\begin{aligned} f_0(z) &= \sum_{\ell=2}^{n} (-1)^{\ell} e_\ell(z) \\ &= \begin{cases} 0, & z = 0, \\ |z| - 1, & z \neq 0, \end{cases} \pmod 4. \end{aligned} \tag{4}$$

where e_ℓ is the elementary symmetric polynomial and $|z|$ is the Hamming weight. The reference measurement has $D_0(z) = i^{f_0(z)}$ and radius

$$c_n = 2^{1-n}, \qquad r_E(n) = \sqrt{3}\, c_n. \tag{5}$$

To compute the seed's signed components, we separate a background state from a correction at the physical zero string. For any real function Θ on $\mathbb{F}_2^n$, set

$$\begin{aligned} |\chi_\Theta\rangle &= 2^{-n/2} \sum_z i^{|z|} e^{i\Theta(z)} |z\rangle, \\ D_\Theta(z) &= \begin{cases} 1, & z = 0, \\ i^{f_0(z)} e^{i\Theta(z)}, & z \neq 0. \end{cases} \end{aligned} \tag{6}$$

The seed state satisfies

$$|\psi_0\rangle = -iC_n \left[|\chi_\Theta\rangle + 2^{-n/2} \big(i - e^{i\Theta(0)}\big) |0^n\rangle \right]. \tag{7}$$

The correction changes one physical input amplitude, irrespective of how Θ is represented.

Lemma 1 (Local phase mechanism)**.** *For a signed local Pauli operator, write*

$$R = C_n^\dagger \widetilde{P} C_n = i^k X^a Z^b, \quad h = a \oplus b, \quad \Gamma = i^{k-|a|}, \tag{8}$$

and denote the background term by $B_R = \langle \chi_\Theta | R | \chi_\Theta \rangle$. Then

$$\langle \widetilde{P} \rangle = B_R + c_n \,\mathrm{Re}\Big[\Gamma e^{-i\Theta(a)} \big(i - e^{i\Theta(0)}\big)\Big]. \tag{9}$$

If all these signed expectations equal $m \geq 0$, every measurement column has a true one-qubit reduced state

$$\rho_{j,e,\sigma} = \frac{1}{2} \left[I + \sqrt{3} m\, \mathbf{v}_{j,\kappa_j,\sigma} \cdot \boldsymbol{\sigma} \right], \tag{10}$$

where $e_0 = e_n = 0$, $\kappa_j = e_{j-1} \oplus e_j$, and

$$\begin{aligned} \mathbf{v}_{j,q,s} &= \frac{1}{\sqrt{3}} \big((-1)^q, \epsilon_j (-1)^{q+s}, (-1)^s\big), \\ \epsilon_j &= \begin{cases} 1, & j < n, \\ -1, & j = n. \end{cases} \end{aligned} \tag{11}$$

For each j, these four unit vectors form a regular tetrahedron, and each label (q, s) occurs 2^{n-2} times.

Every operator $R = C_n^\dagger \widetilde{P} C_n$ that pulls a local output observable $\widetilde{P}$ back to the input of C_n has $a \neq 0$, so the correction in Eq. (7) has no diagonal contribution to a local expectation. Its two cross terms give Eq. (9). Conjugation by the orbit operators in Eq. (3) gives the tetrahedral signs in Eq. (11). Appendix A computes these operators by conjugating successively through the circuit gates and derives the reduced states.

For the Boolean phase $\Theta = tq$, $q(0) = 1$, the pullbacks fall into two classes: class I has $\Gamma = -i$ and $h \cdot a = 1$, while class II has $\Gamma = -1$ and $h \cdot a = 0$. With normalized Walsh coefficients

$$\begin{aligned} W_q(h) &= 2^{-n} \sum_z (-1)^{q(z)\oplus h\cdot z}, \\ D_a q(z) &= q(z) \oplus q(z \oplus a). \end{aligned} \tag{12}$$

the background term is

$$B_R(t) = \begin{cases} -\sin t\, W_q(h), & \text{class I}, \\ -\sin^2(t/2)\, W_{D_a q}(h), & \text{class II}. \end{cases} \tag{13}$$

These formulas express the background terms in terms of Walsh coefficients of the phase function and its derivatives.

III. A UNIFIED FAMILY FOR ALL $n \geq 5$

We now construct a radius-increasing family for every integer $n \geq 5$. Equation (9) suggests two design requirements: cancel every background term $B_R(t)$, and make the remaining cross terms equal. For Boolean phases, the first requirement gives the Walsh cancellations in Eq. (13), while the second fixes the values $q(a)$ at the local flips. For every integer $n \geq 5$, four linear features reduce these requirements to conditions on one fixed four-bit kernel. Define the two internal parity blocks

$$\begin{aligned} E_n &= \{2, 4, \ldots\} \cap [2, n-2], \\ O_n &= \{3, 5, \ldots\} \cap [2, n-2], \end{aligned} \tag{14}$$

and the four linear features

$$\begin{aligned} x &= z_1 \oplus z_{n-1}, & b &= \bigoplus_{j\in E_n} z_j, \\ c &= \bigoplus_{j\in O_n} z_j, & s &= z_1 \oplus z_n. \end{aligned} \tag{15}$$

Write $\mathcal{L}_n z = (x, b, c, s)$. Both internal blocks are nonempty; the four features are independent. On $\mathbb{F}_2^4$, take

$$F(x, b, c, s) = \begin{cases} 1 \oplus (1 \oplus c)(x \oplus b), & s = 0, \\ b \oplus xc, & s = 1. \end{cases} \tag{16}$$

Define

$$q_n(z) = F(\mathcal{L}_n z), \qquad d_n(z) = q_n(z) - \delta_{z,0}. \tag{17}$$

Here the subtraction is in the integers. Since $F(0) = 1$, d_n is still Boolean. Only the physical zero string is removed, not the whole fiber $\mathcal{L}_n z = 0$.

Figure 1(a) shows how the same four-feature kernel determines the phase at every $n \geq 5$; panel (b) displays the resulting radius in units of $r_E(n)$.

Theorem 1 (Unified radius-increasing family)**.** *For every integer $n \geq 5$, let*

$$D_{n,t}(z) = i^{f_0(z)} e^{itd_n(z)}, \qquad M_n(t) = C_n D_{n,t} H^{\otimes n}. \tag{18}$$

For every real t, its columns are an orthonormal basis, and their one-qubit reduced states are Eq. (10) with

$$m_n(t) = c_n(1 - \sin t), \qquad r_n(t) = r_E(n)(1 - \sin t). \tag{19}$$

The tetrahedral directions are independent of t. On

$$J = [-\pi/2, \pi/2], \qquad r_n(J) = [0, 2r_E(n)], \tag{20}$$

the radius is continuous and strictly decreasing. At $t = 0$, the reference measurement is recovered column by column; at $t = -\pi/2$ the radius is $2r_E(n)$, and at $t = \pi/2$ every one-qubit reduced state is $I/2$.

Proof. The proof separates the dimension-dependent feature map from six fixed four-bit cancellations. First, any sum of the form

$$2^{-n} \sum_z (-1)^{h\cdot z} G(\mathcal{L}_n z) \tag{21}$$

vanishes when h is outside the row space of $\mathcal{L}_n$: a translation $z \mapsto z \oplus w$ with $\mathcal{L}_n w = 0$ and $h \cdot w = 1$ reverses each summand's sign. Inside that row space, write $h = \mathcal{L}_n^T H$. Equal fiber sizes reduce Eq. (21) to $2^{-4}\sum_v (-1)^{H\cdot v} G(v)$. The same argument applies to a derivative of q_n, since $D_a q_n = (D_A F) \circ \mathcal{L}_n$ with $A = \mathcal{L}_n a$.

Second, the local characters in the pullback table leave only

$$\begin{aligned} W_F(9) = W_F(11) = W_F(15) = 0, \\ W_{D_6F}(6) = W_{D_7F}(13) = W_{D_7F}(11) = 0. \end{aligned} \tag{22}$$

Integers encode (x, b, c, s) with x the most significant bit; the Walsh normalization is 2^{-4}. The six sums vanish by pairing over feature bits. Appendix A 4 gives the finite calculation and proves that these six identities cover every local component for all $n \geq 5$, including the additional residual terms at $n = 5, 6, 7$. Equation (13) therefore gives $B_R(t) = 0$ identically.

Third, every class-I flip satisfies $F(\mathcal{L}_n a) = 0$, and every class-II flip satisfies $F(\mathcal{L}_n a) = 1$. Equation (9) therefore reduces, respectively, to

$$\begin{aligned} c_n \operatorname{Re}[-i(i - e^{it})] &= c_n(1 - \sin t), \\ c_n \operatorname{Re}[-e^{-it}(i - e^{it})] &= c_n(1 - \sin t). \end{aligned} \tag{23}$$

Lemma 1 transfers this seed identity to all measurement columns. The endpoint, reference-point, and image statements follow from the diagonal phase and the strict monotonicity of $\sin t$ on J. □

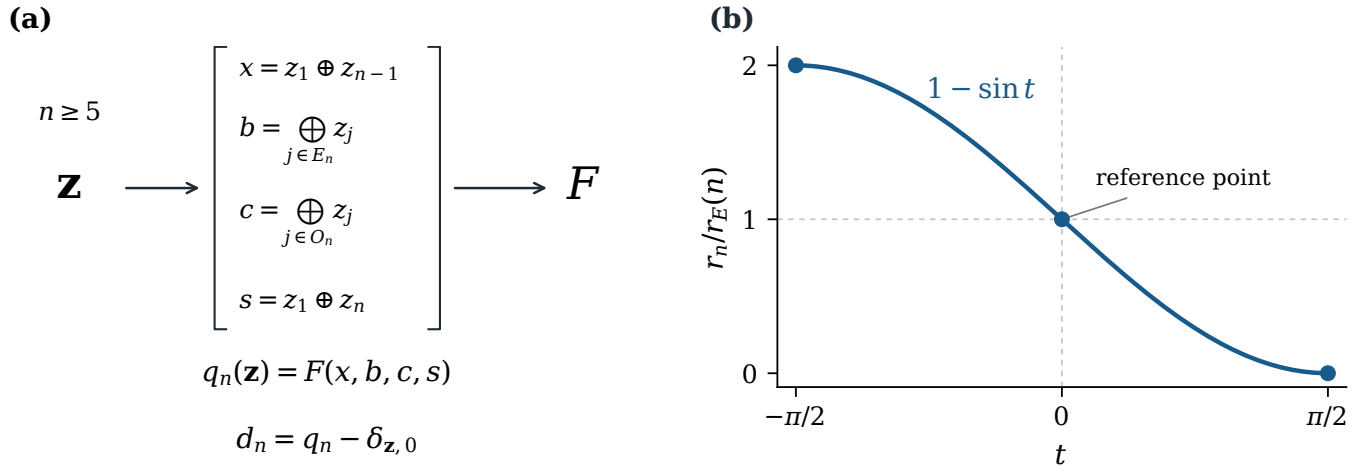


FIG. 1. Unified phase deformation for every $n \geq 5$. (a) Four linear Boolean features feed the same kernel F; the only removed support point is the physical string $\mathbf{z} = 0$. The blocks E_n and O_n contain the even and odd indices in $[2, n-2]$, respectively. (b) The normalized common Bloch radius $r_n(t)/r_E(n) = 1 - \sin t$ on $[-\pi/2, \pi/2]$, where $r_E(n) = \sqrt{3}/2^{n-1}$ is the reference radius. From left to right, the marked points give the endpoint with twice the reference radius, the specified EJM at $t = 0$, and the endpoint with maximally mixed one-qubit marginals.

The increase in radius has a direct interpretation in terms of entanglement between one qubit and the remaining system. All outcomes and all $1|(n-1)$ cuts share the spectrum

$$\lambda_\pm = \frac{1 \pm r_n(t)}{2}, \qquad \mathrm{Tr}\,\rho_j^2 = \frac{1 + r_n(t)^2}{2}. \tag{24}$$

Their entanglement entropy is $h_2((1 + r_n(t))/2)$, where $h_2(p) = -p \log_2 p - (1-p)\log_2(1-p)$. Increasing the Bloch radius decreases this bipartite entanglement entropy. At $t = \pi/2$, the point correction in Eq. (7) vanishes; at $t = -\pi/2$, its interference produces twice the reference local signal. Thus four linear features suffice to express the phase for every $n \geq 5$.

IV. SHARP BOUNDS FOR STRAIGHT BOOLEAN PHASE PATHS

This section classifies the radius laws generated by two-valued real functions and proves that the radius-increasing family attains the largest radius in this broader class. We allow real generators taking at most two values on the physical input strings, including constant generators. Subtracting the generator's value at $z = 0$ removes a global phase. A constant generator then becomes zero; otherwise, rescaling or reversing t makes the two values 0 and 1. We can therefore use a Boolean generator, without assuming a compression structure or vanishing background terms, and write

$$\begin{aligned} M_d(t) &= C_n\,\mathrm{diag}_z\big(i^{f_0(z)} e^{itd(z)}\big) H^{\otimes n}, \\ d &: \mathbb{F}_2^n \longrightarrow \{0,1\}, \qquad d(0) = 0. \end{aligned} \tag{25}$$

We call such a line admissible if, on an open interval containing $t = 0$, its local Pauli orbits form regular tetrahedra with a common radius. This requirement refers to the complete measurement. The unitary factorization already guarantees orthonormality for every real t.

When the signed local components share a value $m_d(t)$, the common Bloch radius is

$$r_d(t) = \sqrt{3}\,|m_d(t)|. \tag{26}$$

Intervals with $m_d \geq 0$ retain the original tetrahedral orientation.

Theorem 2 (Four allowed radius laws and the sharp Boolean-path bound)**.** *Let $n \geq 3$ and let d be admissible. Set $c_n = 2^{1-n}$. All signed local components have one common value $m_d(t)$ for every real t, and, for one fixed pair $(s,k) \in \{0,1\}^2$,*

$$m_d(t) = c_n\big[(1-s) + s\cos t - k \sin t\big]. \tag{27}$$

Thus the four possible functional forms of m_d/c_n are 1, $\cos t$, $1 - \sin t$, and $\cos t - \sin t$. Not every form occurs in every dimension. For every $n \geq 2$, with $r_E(n) = \sqrt{3}\,c_n$, the exact maximum over admissible lines is

$$\max_{\substack{d\ admissible \\ t \in \mathbb{R}}} r_d(t) = \begin{cases} r_E(n), & n = 2,3,4, \\ 2r_E(n), & n \geq 5. \end{cases} \tag{28}$$

Both maxima are attained.

At the reference point all these components equal $c_n > 0$. Regularity therefore makes them equal on a neighborhood of the reference point. Each component is a linear combination of 1, $\cos t$, and $\sin t$, so equality on that neighborhood implies equality for all real t.

The essential restriction comes from comparing coefficients for the two classes of pulled-back Pauli operators.

Put $q = d + \delta_0$, so that q is Boolean and $q(0) = 1$, and use the Walsh coefficients and Boolean derivatives defined in Eq. (12). For class I, the coefficient of $\cos t$ is $c_n q(a)$; for class II, the coefficient of $\sin t$ is $-c_n q(a)$. Since both classes occur for $n \geq 3$, equality of the components forces a common flip value s in class I and a common flip value k in class II. The remaining coefficients agree precisely when

$$\begin{aligned} q(a) &= s, \quad W_q(h) = c_n(s+k-1) \quad \text{(I)}, \\ q(a) &= k, \quad W_{D_a q}(h) = 2c_n(s+k-1) \quad \text{(II)}. \end{aligned} \tag{29}$$

These conditions give Eq. (27) and are also sufficient. For the radius-increasing law $1 - \sin t$, they force both Walsh coefficients to vanish, so Eq. (13) makes the background terms identically zero.

The maximum absolute values of the four normalized functions are, respectively, 1, 1, 2, and $\sqrt{2}$. This proves the upper bound $2r_E$ for every $n \geq 3$. The low-dimensional restrictions are stronger. For $n = 3$, two class-I character equations force $s = k = 0$. For $n = 4$, four such equations sum to $4k = 0$, leaving only the constant and cosine laws. The calculation for $n = 2$, where class II is absent, also permits only these two laws. Appendix B gives the coefficient calculation and shows how the low-dimensional conditions rule out both radius-increasing laws. The reference point attains the low-dimensional maximum, while the unified family attains $2r_E$ at $t = -\pi/2$ for every $n \geq 5$.

Consequently, $n = 5$ is the smallest system size permitting an increase of the radius along a straight phase path through the specified reference point with a two-valued real generator. General multivalued straight phase paths and curved paths require a separate analysis; the case $n = 4$ below makes this distinction concrete.

V. EMBEDDING THE EVEN-QUBIT FAMILY

The unified construction and the earlier even-qubit family provide two phase directions through the same reference measurement. We now ask whether both directions can be varied independently while preserving the tetrahedral geometry. To compare them at the level of the complete measurement, first express the earlier family in the notation of Sec. II. For even n, retain $E_n = \{2, 4, \ldots, n-2\}$ and define

$$q_A(z) = \prod_{j \in E_n} (1 - z_j), \qquad d_A(z) = q_A(z) - \delta_{z,0}. \tag{30}$$

The diagonal $i^{f_0(z)} e^{i\alpha d_A(z)}$ is precisely the known even-qubit family [13], with parameter correspondence $\alpha = \pi\tau/2$. The phase factors in its background terms depend only on the even internal coordinates. Every character h in Table II contains a coordinate outside E_n, so a flip of that coordinate cancels the background expectation. Moreover, $q_A(a) = 1$ for class I and $q_A(a) = 0$ for class II. Equation (9) then gives the common signed component $c_n \cos\alpha$, and hence $r = r_E(n)\cos\alpha$ for $|\alpha| \leq \pi/2$. For $n = 2$, the set E_2 is empty, so the empty-product convention gives $q_A \equiv 1$. Only class I occurs, and its background terms still cancel in pairs. For $n = 4$, $E_4 = \{2\}$ gives $q_A = 1 - z_2$, with the same cancellations and flip values. The cosine law therefore includes both low-dimensional cases.

For even $n \geq 8$, this family and the unified radius-increasing family are coordinate sections of a two-parameter family. Write $q_B = q_n$ and take the Boolean phase generator $d_B = d_n$ from Eq. (17). Set

$$\begin{aligned} D_{\alpha,\beta}(z) &= i^{f_0(z)} e^{i\alpha d_A(z) + i\beta d_B(z)}, \\ M_n(\alpha,\beta) &= C_n D_{\alpha,\beta} H^{\otimes n}. \end{aligned} \tag{31}$$

Figure 2 shows the two-parameter family and its coordinate sections.

Theorem 3 (Two-parameter embedding)**.** *For every even $n \geq 8$, Eq. (31) defines an orthonormal measurement for all real α, β. All its signed seed components equal*

$$m_n(\alpha,\beta) = c_n(\cos\alpha - \sin\beta). \tag{32}$$

On the domain

$$\Omega = \left\{ (\alpha,\beta) : |\alpha| \leq \frac{\pi}{2}, \quad -\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2} - |\alpha| \right\}, \tag{33}$$

the measurement has the fixed local tetrahedra of Eq. (10), with

$$\begin{aligned} r_n(\alpha,\beta) &= r_E(n)(\cos\alpha - \sin\beta), \\ r_n(\Omega) &= [0, 2r_E(n)]. \end{aligned} \tag{34}$$

Its $\beta = 0$ section recovers the earlier even-qubit family column by column, and its $\alpha = 0$ section recovers Theorem 1 column by column. The two parameter directions are independent as ordered measurement projectors.

The exception at $n = 6$ concerns only the two-dimensional extension; the unified radius-increasing family still exists at this size. The direct superposition in Eq. (31) has equal signed components if and only if $\sin\alpha \sin\beta = 0$. Even if the inputs are grouped by the four values of (q_A, q_B), with an independent phase for each group and for the physical zero string, this four-group phase model retains only the two original one-parameter branches near the reference point. Appendix C gives the mixed-term calculation and the local classification.

The essential step in proving Theorem 3 is cancellation of the joint background terms. Put $u = e^{i\alpha}$ and $v = e^{i\beta}$. The seed before applying $-iC_n$ is

$$2^{-n/2} \left[\sum_z i^{|z|} u^{q_A(z)} v^{q_B(z)} |z\rangle + (i - uv)|0^n\rangle \right], \tag{35}$$

At every class-I flip, $(q_A(a), q_B(a)) = (1, 0)$; at every class-II flip, the pair is $(0, 1)$. Their defect contributions therefore agree, because

$$\begin{aligned} \mathrm{Re}[-iu^{-1}(i - uv)] &= \cos\alpha - \sin\beta, \\ \mathrm{Re}[-v^{-1}(i - uv)] &= \cos\alpha - \sin\beta. \end{aligned} \tag{36}$$

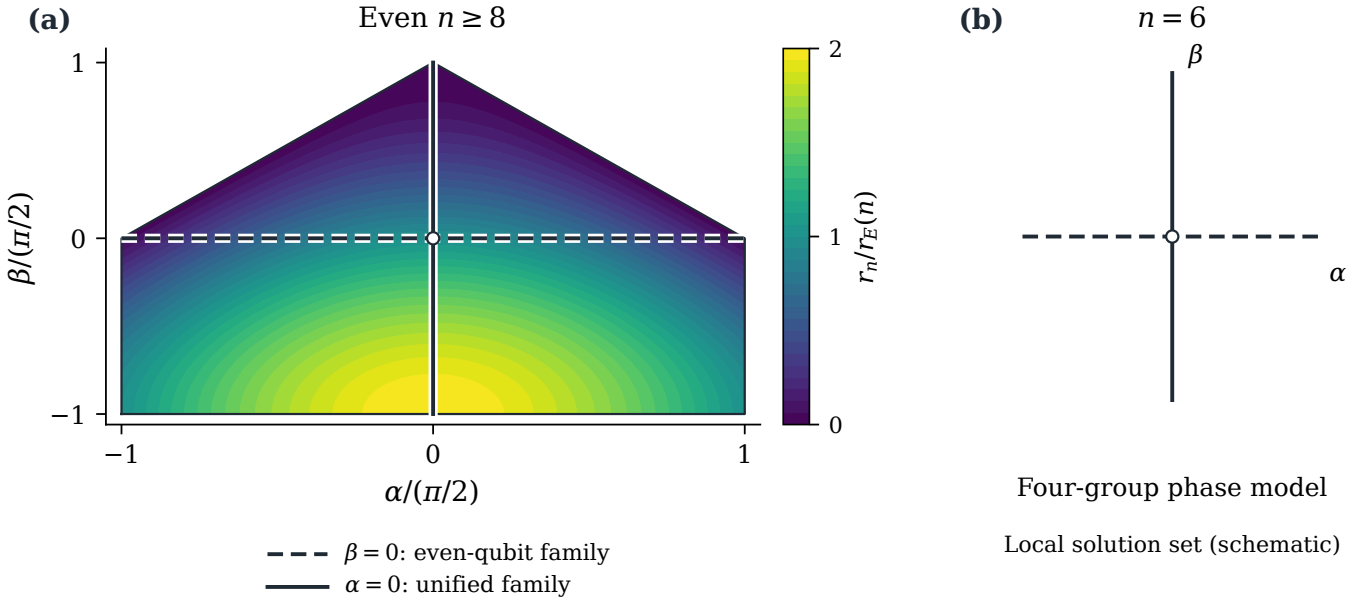


FIG. 2. Two coordinate sections of an even-qubit two-parameter family. (a) For even $n \geq 8$, the domain Ω is given by $|\alpha| \leq \pi/2$ and $-\pi/2 \leq \beta \leq \pi/2 - |\alpha|$. Color represents $r_n/r_E(n) = \cos\alpha - \sin\beta$, with reference radius $r_E(n) = \sqrt{3}/2^{n-1}$; both axes are in units of $\pi/2$. The dashed section $\beta = 0$ recovers the prior even-qubit cosine family, and the solid section $\alpha = 0$ recovers the unified radius-increasing family, in each case column by column. The open circle marks the common reference point. (b) Schematic local solution set at $n = 6$ in the four-group phase model, with phases assigned to the four values of (q_A, q_B) and an independent zero-string phase: the two branches cross at the reference point. This exception concerns the two-dimensional extension; the radius-increasing one-parameter family remains valid at $n = 6$.

Class I leaves q_A unchanged, reducing its background term to the already vanishing q_B term. In class II, the AND difference separates the background sum into the three sectors $q_A(z) - q_A(z \oplus a) = 1, 0, -1$. In the first sector all even internal bits vanish. For $n \geq 8$, there are at least two odd internal bits, and any nonconstant character on this block cancels by a parity-preserving pair flip. Only six fixed four-bit sums remain; their coefficients vanish. Appendix C supplies the full classification and the six finite identities. This establishes Eq. (32) for every even $n \geq 8$, after which the orbit formula gives all reduced states.

On Ω, $\sin\beta \leq \cos\alpha$, while $\cos\alpha - \sin\beta \leq 2$; the $\alpha = 0$ section attains every value between zero and two. The two axis identities follow directly from the diagonal entries in Eq. (31). To check that the parameters survive passage to projectors, note that a diagonal infinitesimal input phase A produces only column phases exactly when $AH^{\otimes n} = H^{\otimes n}B$ for a diagonal B. Every Hadamard entry is nonzero, so A and B must be the same scalar. The pairs (d_A, d_B) take the values $(0,0), (1,0), (0,1)$ at $0, e_1, e_1 \oplus e_2$, respectively. No nonzero linear combination of the two directions is scalar, so none can change only the column phases while leaving all measurement projectors fixed. The two parameter directions are therefore independent at the level of ordered projectors.

VI. LOW-DIMENSIONAL RIGIDITY AND CURVED DEFORMATIONS

This section examines admissible phase variations near the reference point in low dimensions, identifying the restrictions on straight paths and the role of curvature. At $n = 3$ the specified measurement is locally isolated. At $n = 4$, every admissible straight phase path fails to increase the radius, while a curved path through the same reference point does increase it.

Proposition 1 (Local rigidity at $n = 3$)**.** *In the fixed Pauli-orbit representation, the specified EJM at $n = 3$ is isolated in a neighborhood of its phase vector, after removing the overall phase.*

For $n = 3$, removing the overall phase leaves seven independent phase variables. The three qubits have nine signed local components; subtracting the X_1 component from each of the other eight gives eight equality constraints. The Jacobian of seven of these constraints with respect to the seven phase variables has determinant $1/4096$ at the reference point. The inverse function theorem therefore isolates the zero of these seven equations. Since every signed component equals $1/4$ at that point, the squared regularity conditions are equivalent to signed equality in a neighborhood. The Jacobian given in Appendix D 1 turns the numerical rank reported in Ref. [13] into an analytic local statement.

At $n = 4$, straight Boolean phase paths cannot increase the radius. We next ask whether arbitrary real generators can overcome this restriction, writing a general straight phase path as

$$M_d(t) = C_4 \,\mathrm{diag}\big(i^{f_0(z)} e^{itd_z}\big) H^{\otimes 4}, \qquad d_0 = 0. \tag{37}$$

The normalization $d_0 = 0$ removes only a common phase; the remaining 15 entries of d are arbitrary real numbers.

Figure 3 compares the bound for all real straight phase paths with the increase obtained along the curved path below.

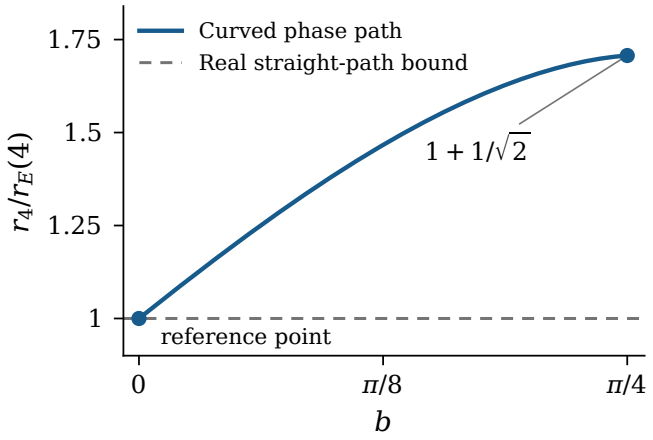


FIG. 3. Radius increase at $n = 4$ along the explicit curved phase path. The solid curve is $r_4/r_E(4) = \cos[b+\delta(b)]-\sin\delta(b)$, evaluated using the closed-form root $\delta(b)$ in Eq. (41) on $0 \le b \le \pi/4$; the reference radius is $r_E(4) = \sqrt{3}/8$. It increases strictly from 1 to $1 + 1/\sqrt{2}$. The dashed level is the radius bound for all admissible real straight phase paths through the specified reference point.

Theorem 4 (Straight-path bound and a curved path at $n = 4$)**.** *Every line* (37) *that remains a common-radius regular tetrahedral measurement in a neighborhood of* $t = 0$ *has*

$$r_d(t) = r_E(4)\,|\cos(\omega t)| \le r_E(4) \tag{38}$$

for some real ω*. Nevertheless, there is an explicit curved phase path through the same measurement for which*

$$r_4\big([0,\pi/4]\big) = \left[r_E(4), \left(1 + \frac{1}{\sqrt{2}}\right) r_E(4)\right], \tag{39}$$

and the radius is strictly increasing along the path.

The straight-path bound follows by expanding the component-equality conditions at the reference point. First-order conditions restrict the fifteen-dimensional space of real phase directions to a four-dimensional subspace. Requiring the same straight path to satisfy the second- and third-order conditions reduces the admissible directions to two two-dimensional subspaces. Substitution into the full formulas then gives the cosine law (38) on both. Appendix D 2 provides the classification and verification. This restricts paths with a fixed direction, while allowing curved paths whose direction varies with the parameter.

To construct such a curved path, first introduce two independent real parameters b, δ and set $\theta_z = bu_z + \delta v_z$, using the two fixed integer phase directions u, v in Eq. (D14). For these phases, the signed physical Pauli components of the seed take only two values: ten components equal $T/8$, whereas the components of X_1 and $-Y_4$ equal $H/8$, where

$$\begin{aligned} T &= \cos(b+\delta) - \sin\delta, \\ H &= \cos b + 2\sin\delta\cos b + 4\sin b\cos b. \end{aligned} \tag{40}$$

These identities hold before a path is selected. The signed regularity condition is therefore $H = T$, a quadratic equation in $\tan(\delta/2)$.

The root branch through $(b, \delta) = (0, 0)$ has the following closed expression. For $0 \le b \le \pi/4$, define

$$\begin{aligned} A(b) &= \cos b(1 + 2\sin b), \\ B(b) &= 2\cos b + \sin b + 1, \qquad C(b) = 2\sin b\cos b, \\ \delta(b) &= -2\arctan\frac{2C(b)}{B(b) + \sqrt{B(b)^2 - 4A(b)C(b)}}. \end{aligned} \tag{41}$$

With this choice, the phase path and measurement are

$$\begin{aligned} \theta_z(b) &= bu_z + \delta(b)v_z, \\ M_4(b) &= C_4\,\mathrm{diag}\big(i^{f_0(z)} e^{i\theta_z(b)}\big) H^{\otimes 4}. \end{aligned} \tag{42}$$

Thus all signed components coincide on the entire explicit path. Unitarity and the Pauli-orbit relation then give the complete measurement and its fixed local tetrahedra. Its common radius is

$$r_4(b) = r_E(4)\big[\cos\big(b + \delta(b)\big) - \sin\delta(b)\big]. \tag{43}$$

The radicand and denominator in Eq. (41) are strictly positive on the closed interval $0 \le b \le \pi/4$. Appendix D 3 proves that the branch is well defined and that its radius is strictly increasing throughout this interval. Both angles vanish at $b = 0$, so the original measurement is recovered column by column. At the other endpoint $\delta(\pi/4) = -\pi/4$, giving the upper endpoint in Eq. (39). The curved path leaves the two planes of admissible line directions and has $r_4'(0) = \sqrt{3}/6 > 0$.

The larger value $3r_E(4)$ already occurs at a different measurement at $n = 4$ in Ref. [13]. The purpose of Eq. (42) is to increase the radius through the specified f_0 measurement and explain the required curvature. The parameter segment of that earlier family contains no measurement equivalent to this specified point under local unitaries, qubit permutations, and outcome relabeling. Appendix D 4 proves this inequivalence using the one-qubit spectrum and the sum of the three $2|2$ purities. Whether a larger regular-tetrahedral solution manifold connects the two families remains open.

TABLE I. Deformations through the specified reference EJM in the fixed phase representation. Among all admissible straight Boolean phase paths, the maximum radius is $r_E(n)$ for $n = 2, 3, 4$ and $2r_E(n)$ for $n \geq 5$, where $r_E(n) = \sqrt{3}/2^{n-1}$ is the reference radius. At $n = 4$, a curved path increases the radius despite the bound on all real straight phase paths. For even $n \geq 8$, the radius-increasing and cosine families belong to one two-parameter family; at $n = 6$, the stated two-dimensional extension fails while the one-parameter family remains valid.

Qubits	Boolean-path maximum	Further deformation structure	Relation to the even family
$n = 2$	$r_E(2)$	Known cosine family	Empty-product case of q_A
$n = 3$	$r_E(3)$	Reference point locally isolated	—
$n = 4$	$r_E(4)$	All real straight paths bounded; a curved path increases the radius	$q_A = 1 - z_2$
$n \geq 5$	$2r_E(n)$	One four-feature kernel for every n	Two-dimensional embedding for even $n \geq 8$ At $n = 6$: two crossing branches in the four-group phase model

VII. DISCUSSION

This section discusses the mechanism of the unified construction and how the shape of a phase path and the compatibility of different deformation directions constrain radius tuning. Table I summarizes the radius bounds and deformation structures at different system sizes.

The phase representation treats global orthogonality and local geometric constraints separately: changes in the diagonal phases always preserve the orthonormality of the complete measurement basis, while the local regular-tetrahedral conditions are determined by relations among the signed Pauli expectation values. Equation (9) further decomposes these expectation values into background terms and interference terms involving the correction at the physical zero string. The four-feature kernel of the unified construction cancels all background terms, leaving the interference terms to control local polarization and reducing the calculations at different system sizes to fixed-size cancellation identities. This explains why one construction applies to both odd and even numbers of qubits. The continuously varying common Bloch radius also gives all measurement basis states the same entanglement-tuning law across every single-qubit-versus-rest cut: as the radius increases, the single-qubit purity increases and the corresponding bipartite entanglement entropy decreases.

The ability to increase the radius also depends on the shape of the phase path. Among straight phase paths with two-valued generators through the specified reference measurement, the classification of the allowed radius laws gives the attainable sharp radius bound and identifies $n = 5$ as the smallest system size allowing a radius increase. The four-qubit result further shows that this restriction does not arise solely from the two-valued nature of the generators: even with arbitrary real generators, no admissible straight line increases the reference radius, whereas an explicit curved path through the same reference point does. Thus a classification of admissible straight lines does not exhaust the possible continuous deformations. The nonlinear coordination of phases needed to preserve the local geometry is central to the four-qubit construction that increases the radius.

Compatibility between different deformation directions imposes a further constraint. For even $n \geq 8$, the new radius-increasing family and the known cosine family embed into a common two-dimensional measurement family. Their joint background terms cancel through parity-preserving pair flips and finite identities. At six qubits, the relevant odd internal block contains only one position, and direct superposition leaves differences between local components proportional to $\sin\alpha \sin\beta$. Even when the four input classes are assigned independent phases and the zero-string phase is varied separately, this model retains only the two original branches near the reference point. Two deformations being individually admissible therefore does not suffice to make them independently tunable parameters of one family.

These results raise two questions about more general phase solution sets. First, at four qubits, Ref. [13] gives another measurement family reaching radius $3r_E(4)$. Appendix D 4 proves that its specified parameter segment contains no measurement equivalent to our reference measurement under local unitaries, qubit permutations, and outcome relabeling. This does not determine the connectivity of the two families. Within the fixed phase representation, can the curved branch constructed here be connected to the known larger-radius family by a continuous path preserving regular tetrahedra with a common radius? This question concerns the measurements continuously reachable from the specified reference measurement and whether the two radius-increasing constructions belong to the same connected component.

Second, at six qubits, the requirement of a common phase within each group could be relaxed by assigning different additional phases to computational-basis components with the same (q_A, q_B) label. Whether these additional degrees of freedom can compensate for the local-component differences generated by the two deformations acting together, and produce a two-dimensional measurement family containing both known branches, remains open. Resolving this question would determine

whether the observed crossing is a consequence of requiring a common phase within each group or a compatibility obstruction that persists under more general phase deformations.

VIII. CONCLUSIONS

We have constructed a continuous family of elegant joint measurements for every integer $n \geq 5$, with fixed local tetrahedral directions and common radius covering $[0, 2r_E(n)]$ through the specified reference measurement. The upper endpoint is attained and optimal among all admissible straight phase paths with two-valued real generators in the fixed phase representation. A single four-feature mechanism thus gives continuous control of local polarization while preserving global orthogonality. The embedding of the radius-increasing and known cosine families for even $n \geq 8$, together with the radius increase along a curved path at $n = 4$, further shows how compatible phase directions and the shape of the phase path determine the available deformations.

The main mathematical results of this paper have been formally proved in Lean.

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We used OpenAI Codex (GPT-6 Astra) to assist in translating the mathematical proofs in this paper into Lean code and to check the manuscript for grammatical errors. All AI outputs were reviewed by the authors.

DATA AND CODE AVAILABILITY

The Lean formalization source code, dependency configuration, and reproduction instructions supporting the results of this paper are archived on Zenodo at DOI: 10.5281/zenodo.23882488.

Appendix A: Local formulas and the all-n reduction

This appendix derives the local phase formula in Lemma 1 and proves the all-n cancellations and orbit statements used in Theorem 1.

1. Complete measurement, reduced states, and multiplicity

This section proves that the seed's signed local components determine the reduced states and tetrahedral multiplicities of the complete measurement. The circuit obeys

$$\begin{aligned} C_n^\dagger (Z_h Z_{h+1}) C_n &= Z_h \quad (h < n), \\ C_n^\dagger X^{\otimes n} C_n &= Z_n. \end{aligned} \tag{A1}$$

Since D is diagonal and $H^{\otimes n}|k\rangle = Z^k|+\rangle^{\otimes n}$, these identities prove Eq. (3) with the original column ordering $k = (e_1, \ldots, e_{n-1}, \sigma)$. No parameter-dependent relabeling or column phase is needed. At site j, Pauli conjugation by the orbit element multiplies the physical X, Y, Z expectations by $(-1)^{\kappa_j}, (-1)^{\kappa_j+\sigma}, (-1)^{\sigma}$, respectively. With the seed's orientation ϵ_j, this gives Eq. (11).

For the actual partial trace $\rho_j = \mathrm{Tr}_{\bar{j}}|\psi\rangle\langle\psi|$, one has $\mathrm{Tr}(\rho_j \sigma_\alpha) = \langle\psi|\sigma_{j\alpha}|\psi\rangle$. Normalization and the three Pauli components determine a 2×2 Hermitian matrix uniquely, proving Eq. (10). Each vector in Eq. (11) has norm one. Two distinct labels change exactly two coordinate signs, so their dot product is $-1/3$.

For fixed j, $\kappa_j = e_{j-1} \oplus e_j$ is a nonzero linear functional of e, including $j = 1, n$. It is independent of the column bit σ. Consequently each pair (κ_j, σ) has exactly 2^{n-2} preimages. At positive radius these are four distinct reduced states; at zero radius they all equal $I/2$. Finally, unitarity of $M[D]$ proves both orthonormality and $\sum_k |\psi_k\rangle\langle\psi_k| = I$. Thus the local geometry of the full measurement reduces to the signed components of its seed.

2. Reference phases, operator pullbacks, and the physical point correction

This section derives the seed decomposition and separates each local expectation into a background term and the interference with the physical zero-string correction.

The reference-phase identity follows from

$$\sum_{\ell=0}^{w} (-1)^\ell \binom{w}{\ell} = 0 \quad (w > 0). \tag{A2}$$

Removing the $\ell = 0, 1$ terms gives $f_0 = w - 1$ for $w > 0$. At $z = 0$, the bracket in Eq. (7) has amplitude $2^{-n/2} i$; multiplication by $-i$ restores the reference amplitude $2^{-n/2}$. At every other input it has amplitude $2^{-n/2} i^{f_0(z)} e^{i\Theta(z)}$. This proves the decomposition at the level of the actual input amplitudes.

Let $\mathbf{e}_j$ denote a binary unit vector and $T_j = \mathbf{e}_j \oplus \cdots \oplus \mathbf{e}_n$. Direct conjugation through the CNOT chain gives

$$\begin{aligned} S^\dagger X_1 S &= X_1, \\ S^\dagger X_j S &= X_{j-1} X_j \quad (j > 1), \\ S^\dagger Z_j S &= Z_j Z_{j+1} \cdots Z_n. \end{aligned} \tag{A3}$$

Conjugating by the final-qubit Hadamard and using $Y = iXZ$ gives Table II. In $R = i^k X^a Z^b$, the Z factors act first. For example, Y_n pulls back to $-iX_{n-1}X_nZ_n$; the signed observable $-Y_n$ therefore has $\Gamma = -i$, as do the other class-I rows. Every row has $a \neq 0$ and $h \neq 0$.

TABLE II. Pullbacks $C_n^\dagger \widetilde{P} C_n = i^k X^a Z^b$, expressed by a, $h = a \oplus b$, and $\Gamma = i^{k-|a|}$. Here $\mathbf{e}_j$ is a binary unit vector and $T_j = \mathbf{e}_j \oplus \cdots \oplus \mathbf{e}_n$. The table is valid for every $n \geq 2$; internal rows are absent at $n = 2$.

Class	Signed observable	a	h	Γ
I	X_1	$\mathbf{e}_1$	$\mathbf{e}_1$	$-i$
I	Y_1	$\mathbf{e}_1 \oplus \mathbf{e}_n$	T_2	$-i$
I	Z_j, $1 \leq j \leq n$	$\mathbf{e}_n$	T_j	$-i$
I	X_n	$\mathbf{e}_{n-1}$	T_{n-1}	$-i$
I	$-Y_n$	$\mathbf{e}_{n-1} \oplus \mathbf{e}_n$	$\mathbf{e}_{n-1}$	$-i$
II	X_j, $2 \leq j < n$	$\mathbf{e}_{j-1} \oplus \mathbf{e}_j$	$\mathbf{e}_{j-1} \oplus \mathbf{e}_j$	-1
II	Y_j, $2 \leq j < n$	$\mathbf{e}_{j-1} \oplus \mathbf{e}_j \oplus \mathbf{e}_n$	$\mathbf{e}_{j-1} \oplus T_{j+1}$	-1

For any real Θ, direct summation yields

$$B_R = 2^{-n}\Gamma \sum_z (-1)^{h\cdot z} e^{i[\Theta(z)-\Theta(z\oplus a)]}. \tag{A4}$$

Indeed, $i^{|z|-|z\oplus a|} = i^{-|a|}(-1)^{a\cdot z}$. Also,

$$\langle \chi_\Theta | R | 0 \rangle = 2^{-n/2}\Gamma e^{-i\Theta(a)}, \qquad \langle 0|R|0\rangle = 0. \tag{A5}$$

Expanding Eq. (7) therefore proves Eq. (9), including its normalization and its background term.

3. Boolean backgrounds and character pairing

This section evaluates the Boolean background terms and reduces their character sums to the features on which the phase depends. Put $\xi_z = 2^{-n/2} i^{|z|}$ and $\eta_z = (-1)^{q(z)}\xi_z$. Since q is Boolean,

$$|\chi_{tq}\rangle = e^{it/2}\left[\cos(t/2)|\xi\rangle - i\sin(t/2)|\eta\rangle\right]. \tag{A6}$$

Equation (A4) gives

$$\begin{aligned} \langle \xi|R|\xi\rangle &= 0, \\ \langle \xi|R|\eta\rangle &= \Gamma W_q(h), \\ \langle \eta|R|\eta\rangle &= \Gamma W_{D_a q}(h). \end{aligned} \tag{A7}$$

In class I, $D_a q$ is invariant under translation by a, whereas $h \cdot a = 1$. Its Walsh coefficient vanishes by pairing. The remaining cross term is $-\sin t\, W_q(h)$. In class II, $\Gamma = -1$ is real, so the cross term vanishes and the diagonal term is $-\sin^2(t/2) W_{D_a q}(h)$. This proves Eq. (13) without requiring the background to vanish.

More generally, let $L : \mathbb{F}_2^n \to \mathbb{F}_2^k$ have rank k. If $h \notin \operatorname{row} L = (\ker L)^\perp$, there is a $w \in \ker L$ with $h \cdot w = 1$. Translation by w changes the sign of $(-1)^{h\cdot z}$ and leaves every function of Lz unchanged. If $h = L^T H$, instead,

$$2^{-n}\sum_z (-1)^{h\cdot z} G(Lz) = 2^{-k}\sum_v (-1)^{H\cdot v} G(v), \tag{A8}$$

because every fiber has 2^{n-k} elements. This elementary pairing is the only general linear-encoding fact needed here.

TABLE III. All local terms not canceled by character pairing for the four-feature map. $A = \mathcal{L}_n a$, $h = \mathcal{L}_n^T H$. The integers encode four-bit vectors in the order (x, b, c, s), with x the most significant bit. All rows absent from this table cancel before any property of F is used.

Observable	Dimension	Class	(A, H)
Y_1	$n \geq 5$	I	$(8, 15)$
Z_2	$n \geq 5$	I	$(1, 15)$
Z_{n-1}	$n \geq 5$	I	$(1, 9)$
X_n	$n \geq 5$	I	$(8, 9)$
Z_3	$n = 5$	I	$(1, 11)$
X_3	$n = 5$	II	$(6, 6)$
Y_3	$n = 5, 6$	II	$(7, 13)$
Y_4	$n = 6, 7$	II	$(7, 11)$

4. Verification of the four-feature kernel for every $n \geq 5$

This section verifies background cancellation and the interference terms for the four-feature kernel at every $n \geq 5$.

a. Classification of characters not canceled by pairing. For the four-feature map in Eq. (15), a physical character h lies in its row space exactly when

$$\begin{gathered} h \text{ is constant on } E_n \text{ and on } O_n, \\ h_1 = h_{n-1} \oplus h_n. \end{gathered} \tag{A9}$$

Its four-bit representative is

$$H = (h_{n-1}, h_E, h_O, h_n).$$

The first condition tests the two internal parity features; the second follows because the coefficient of z_1 is the sum of the coefficients of x and s. Conversely these conditions reconstruct the character from the four features. Both internal blocks contain a coordinate, so this representation is unique.

Applying Eq. (A9) to Table II gives exactly Table III. Here and below, a four-bit integer uses (x, b, c, s) in that order.

The classification follows directly from the supports of the characters. For class I, $X_1, Z_1, Z_n, -Y_n$ fail the endpoint relation in Eq. (A9). The four dimension-independent rows in Table III satisfy it and select complete parity blocks. An internal suffix T_j, $3 \le j \le n-2$, can select complete blocks only for $n = 5, j = 3$. For $n \ge 6$, $j = 3$ selects 4 and omits 2; for $j \ge 4$, the selected internal coordinate $n-2$ has the same parity as an omitted first block coordinate 2 or 3.

For class-II X_j, the endpoint cases $j = 2, n-1$ fail the endpoint relation. Otherwise its character selects two neighboring internal coordinates. Both parity blocks can be fully selected only when each has one member, which gives $n = 5, j = 3$. For class-II Y_j, the support is $\{j-1\} \cup \{j+1, \ldots, n\}$. Again $j = 2, n-1$ fail the endpoint relation. At $j = 3$, the odd block must consist only of 3, giving $n = 5, 6$. At $j = 4$, the even block can contain no 6, giving $n = 6, 7$. For $j \ge 5$, the selected coordinate $j-1$ and the omitted first coordinate of the same parity violate block constancy. This proves the table for unbounded n.

b. Cancellation of the six background coefficients. The kernel has the cubic Boolean expansion $F = 1 \oplus x \oplus b \oplus s \oplus xs \oplus xc \oplus bc \oplus bcs$.

To evaluate the terms not canceled by pairing, the coefficient of x in F is $1 \oplus s \oplus c$. All three characters $9, 11, 15$ have x coefficient one, so summing over x leaves only $c = s$. On the slices $c = s = 0$ and $c = s = 1$, respectively,

$$F = 1 \oplus x \oplus b, \qquad F = x \oplus b. \tag{A10}$$

Characters 9 and 11 omit b, so their sums cancel over b. Character 15 contributes $x \oplus b \oplus c \oplus s = x \oplus b$ on these slices, leaving opposite constant signs. These are the first three identities in Eq. (22). Expanding $F(v) \oplus F(v \oplus A)$ gives

$$\begin{aligned} D_6 F &= x \oplus b \oplus c \oplus s(b \oplus c \oplus 1), \\ D_7 F &= bc \oplus bs \oplus cs \oplus s. \end{aligned} \tag{A11}$$

In the first line, x occurs only in the derivative; in the second, it occurs only in characters 13 and 11. Pairing over x proves the remaining three identities. Thus all backgrounds vanish for all real t.

c. Values at every local flip. The character classification concerns the background terms; the interference contribution still requires the kernel value at every flip, including rows whose characters have already canceled. From Table II, class-I flips map into $\{1, 8, 9\}$, while class-II flips map into $\{6, 7, 10, 11, 12, 13\}$. Direct substitution in the two branches of Eq. (16) gives

$$\begin{aligned} F(A) &= 0 \quad (A \in \{1, 8, 9\}), \\ F(A) &= 1 \quad (A \in \{6, 7, 10, 11, 12, 13\}). \end{aligned} \tag{A12}$$

These sets include both parities of the final internal coordinate. They prove the cross-term identities used in Theorem 1.

As supplementary information, the kernel equals one at six inputs with $s = 0$ and four with $s = 1$. Each feature fiber has size 2^{n-4}, and the physical zero string is removed from d_n, so $|\operatorname{supp} d_n| = 10 \cdot 2^{n-4} - 1$.

Appendix B: Proof of the sharp bounds for Boolean straight phase paths

This appendix proves Theorem 2: tetrahedral regularity fixes the signed local components, coefficient comparison gives the four radius laws, and low-dimensional character identities determine which laws are possible.

1. From tetrahedral regularity to equal signed components

This section proves that local tetrahedral regularity near the reference point is equivalent to equality of the signed components along the entire Boolean straight phase path. The local action of the Pauli orbit sends a seed Bloch vector (x, y, z) to the four vectors

$$(x, y, z), \quad (-x, -y, z), \quad (x, -y, -z), \quad (-x, y, -z). \tag{B1}$$

The label map is onto these four sign patterns. The three distinct off-diagonal inner products are $-x^2 - y^2 + z^2$, $x^2 - y^2 - z^2$, and $-x^2 + y^2 - z^2$. They are equal precisely when $x^2 = y^2 = z^2$. A common radius at all sites then makes the squares of all $3n$ signed components equal. At the specified reference point each signed component equals $c_n > 0$. By continuity their signs remain positive on a sufficiently small interval, and the components themselves are equal there.

For Boolean d, every term in a local expectation has a phase $e^{it[d(z) - d(z')]}$ with $d(z) - d(z') \in \{0, \pm 1\}$. Thus any difference of two signed components is $A + B\cos t + C\sin t$. If this vanishes near zero, its value and its first two derivatives at zero give $A = B = C = 0$. Equality therefore holds for all real t, including any zero-radius points. Conversely, equal signed components give the regular orbit in Eq. (B1), with radius $\sqrt{3}\,|m_d|$. This proves the local-to-global step used in Theorem 2.

2. Background terms and coefficient comparison

This section compares the coefficients of the local components to derive the necessary and sufficient conditions and the four radius laws. Write $N = 2^n$, $c = 2/N$, and $q = d + \delta_0$. Then q is Boolean with $q(0) = 1$, and Eq. (7) applies with $\Theta = tq$. Combining the local formula (9) with the Boolean backgrounds in Eq. (13) gives the components for classes I and II of Table II:

$$m_I(t) = \begin{cases} c\cos t - W_q(h)\sin t, & q(a) = 1, \\ c - [c + W_q(h)]\sin t, & q(a) = 0, \end{cases} \tag{B2}$$

TABLE IV. Coefficients of the signed local components after the flip values have been fixed to s in class I and k in class II. Here $W = W_q(h)$ for a class-I row and $V = W_{D_a q}(h)$ for a class-II row. Coefficient matching determines W and V.

Class	Constant	$\cos t$	$\sin t$
I	$c(1-s)$	cs	$-c(1-s)-W$
II	$ck - V/2$	$c(1-k)+V/2$	$-ck$

and

$$\begin{aligned} m_{II}(t) = & -\tfrac{1}{2}W_{D_a q}(h)(1-\cos t) \\ & + c\big\{[1-q(a)]\cos t + q(a)(1-\sin t)\big\}. \end{aligned} \tag{B3}$$

For $n \geq 3$, both classes occur. The cosine coefficient in Eq. (B2) forces all class-I flip values to equal some $s \in \{0,1\}$. The sine coefficient in Eq. (B3) forces all class-II flip values to equal some $k \in \{0,1\}$. Write $W = W_q(h)$ for each class-I row and $V = W_{D_a q}(h)$ for each class-II row. Table IV displays the coefficients to be compared. Equality of the sine and constant coefficients, respectively, gives

$$W = c(s+k-1), \qquad V = 2c(s+k-1). \tag{B4}$$

The cosine coefficients then agree as well. Substitution proves both directions of Eq. (29) and yields the four allowed radius laws. Their maximum absolute values prove the general upper bound.

3. Analytic exclusions at $n = 2, 3, 4$

This section excludes radius increase for $n = 2, 3, 4$ and identifies measurements attaining the bounds in every dimension. Use binary integers with z_1 the most significant bit. For $n = 2$, only class I occurs and its flip masks cover all three nonzero strings. Equality of the cosine coefficients forces $q(z) = s$ for every $z \neq 0$. The choices $s = 0$ and $s = 1$ give $q = \delta_0$ and $q \equiv 1$, respectively. Their common components are c and $c\cos t$, so neither exceeds the base radius in absolute value.

For $n \geq 3$ and every class-I character, the Walsh condition is equivalently

$$\sum_z q(z)(-1)^{h\cdot z} = 1 - s - k. \tag{B5}$$

Indeed, $W_q(h) = -2N^{-1}\sum_z q(z)(-1)^{h\cdot z}$ for $h \neq 0$.

For $n = 3$, the class-I flip masks are $\{1,2,3,4,5\}$ and the class-II flip masks are $\{6,7\}$. Thus $q(0) = 1$, the first set has value s, and the second set has value k; there are no further unknowns. For $h = 1$, the left side of Eq. (B5) is $1 - s$, which gives $k = 0$. For $h = 4$, it is $1 + s - 2k$, which gives $2s - k = 0$. Hence $s = k = 0$ and $d = 0$.

For $n = 4$, the class-I flip masks are $\{1,2,3,8,9\}$ and the class-II masks are $\{6,7,12,13\}$. Put

$$(A,B,C,D,E,F) = (q_4, q_5, q_{10}, q_{11}, q_{14}, q_{15}), \tag{B6}$$

where $q_j = q(j)$. The four class-I characters $h = 1, 3, 7, 15$ give, respectively,

$$\begin{aligned} A - B + C - D + E - F + k &= 0, \\ A - B - C + D - E + F + k &= 0, \\ -A + B - C + D + E - F + k &= 0, \\ -A + B + C - D - E + F + k &= 0. \end{aligned} \tag{B7}$$

Adding these equations gives $4k = 0$ and therefore $k = 0$. Equation (27) now leaves only $m = c$ or $m = c\cos t$. This exclusion covers every Boolean function at $n = 4$; no ansatz for its support was used. It is sufficient for the radius bound, so a list of individual admissible supports is unnecessary.

The low-dimensional upper bounds are attained at $t = 0$. For every $n \geq 5$, the explicit unified generator satisfies the admissibility conditions and gives $m(-\pi/2) = 2c_n$. These two observations complete the proof of Eq. (28), with a maximum rather than merely a supremum.

Appendix C: Mixed backgrounds and the exception at $n = 6$

This appendix proves Theorem 3 and the local $n = 6$ branch statement in Sec. V. It evaluates the mixed backgrounds for even $n \geq 8$ and solves the phase equations for the four input groups near the reference measurement at $n = 6$.

1. Reduction to six finite identities

We prove the joint-background cancellation used in Theorem 3. In this appendix, write $\mathcal{B}_R = 2^n B_R$ for the unnormalized background expectation, where B_R is the normalized quantity in Eq. (9). For a pulled-back signed Pauli $R = i^p X^a Z^b$, put $h = a \oplus b$ and $\Gamma = i^{p-|a|}$. Equation (35) gives

$$\mathcal{B}_R = \Gamma \sum_z (-1)^{h\cdot z} u^{q_A(z) - q_A(z\oplus a)} v^{q_B(z) - q_B(z\oplus a)}. \tag{C1}$$

The sign follows from $i^{|z|-|z\oplus a|} = i^{-|a|}(-1)^{a\cdot z}$.

For class I, a contains no even internal position, so the exponent of u vanishes. The remaining q_B expectation is zero by the unified-family calculation. Every class-II flip contains exactly one position $e \in E_n$. The u exponents 1 and -1 occur precisely on $z|_{E_n} = 0$ and $z|_{E_n} = e_e$, respectively. Write

$$\mathcal{B}_R = \Gamma\left[uK_+(v) + K_0(v) + u^{-1}K_-(v)\right]. \tag{C2}$$

For class II, $h \cdot a = 0$. Translation by a exchanges the two nonzero sectors without changing the character and reverses the v exponent. Hence $K_-(v) = K_+(v^{-1})$. At $u = 1$, the single-q_B result gives $K_+ + K_0 + K_- = 0$. It is therefore enough to prove $K_+ = 0$.

On the positive sector, q_B depends only on

$$w = (z_1, c, z_{n-1}, z_n), \qquad c = \bigoplus_{j\in O_n} z_j,$$
$$O_n = \{3, 5, \ldots, n-3\}. \tag{C3}$$

If h takes both values zero and one on O_n, choose two such positions. Flipping both preserves c, $q_B(z)$, and $q_B(z \oplus a)$, but reverses $(-1)^{h\cdot z}$. The corresponding terms cancel.

The following classification covers all internal observables when $n \geq 8$. For X_j, $h = e_{j-1} \oplus e_j$. Only $j = 2, n-1$ select no odd internal position; every other j selects exactly one of the at least two positions in O_n. For Y_j, $h = e_{j-1} \oplus \bigoplus_{k=j+1}^{n} e_k$. At $j = 2, 4$, its restriction to O_n is all ones. At even $j \geq 6$, it contains $j-1$ but omits 3. At odd $j \leq n-5$, it is a nonempty proper suffix of O_n. At $j = n-3, n-1$, it is zero. Consequently, only

$$X_2,\ X_{n-1},\ Y_2,\ Y_4,\ Y_{n-3},\ Y_{n-1} \tag{C4}$$

remain after character pairing.

We evaluate these six terms by finite counting, using only the four-feature kernel F of Eq. (16). For $w = (w_1, w_2, w_3, w_4)$ define

$$T_\varepsilon(w) = F(w_1 \oplus w_3, \varepsilon, w_2, w_1 \oplus w_4), \qquad \varepsilon \in \{0, 1\}. \tag{C5}$$

Let A encode the flip induced by a on w and let H encode the remaining character. Both are written as four-bit integers, with w_1 the most significant bit. Then

$$K_+(v) = 2^{|O_n|-1} \sum_{w\in\mathbb{F}_2^4} (-1)^{H\cdot w} v^{T_0(w)-T_1(w\oplus A)}. \tag{C6}$$

For $k = -1, 0, 1$, let $N_k^\pm$ count the inputs with exponent k and character sign ± 1. Grouping Eq. (C6) by exponent gives

$$K_+(v) = 2^{|O_n|-1} \sum_{k=-1}^{1} (N_k^+ - N_k^-) v^k. \tag{C7}$$

Table V gives the complete counts obtained by substituting the two branches of F. In every row $N_k^+ = N_k^-$ for each k, so Eq. (C7) vanishes for every v. Equations (C2) and (36) give the common signed component $2^{1-n}(\cos\alpha - \sin\beta)$ for all even $n \geq 8$. Unitarity and the orbit transfer in Eq. (10) then establish the full measurement statement.

2. The direct superposition at $n = 6$

This section determines why direct superposition leaves a mixed background at $n = 6$. Here $O_n = \{3\}$. The two additional terms X_3, X_4 both have $(A, H) = (4, 4)$ in Eq. (C6). Their counts for $k = -1, 0, 1$ are $(4, 2), (4, 4), (0, 2)$; Eq. (C7) therefore gives

$$K_+(v) = 2v^{-1} - 2v,$$
$$K_-(v) = -K_+(v), \qquad K_0(v) = 0. \tag{C8}$$

TABLE V. Positive-sector counts for even $n \geq 8$. Each entry (N_k^+, N_k^-) counts inputs $w \in \mathbb{F}_2^4$ with exponent $k = T_0(w) - T_1(w \oplus A)$ and character sign $(-1)^{H\cdot w} = \pm 1$, respectively. The flip A and character H are four-bit integers in the coordinate order $w = (z_1, c, z_{n-1}, z_n)$, with w_1 the most significant bit. Equal positive and negative counts make every listed sector sum vanish.

Observable	A	H	$k=-1$	$k=0$	$k=1$
X_2	8	8	$(3,3)$	$(4,4)$	$(1,1)$
X_{n-1}	2	2	$(2,2)$	$(6,6)$	$(0,0)$
Y_2	9	15	$(2,2)$	$(6,6)$	$(0,0)$
Y_4	5	7	$(2,2)$	$(6,6)$	$(0,0)$
Y_{n-3}	5	3	$(2,2)$	$(6,6)$	$(0,0)$
Y_{n-1}	3	1	$(3,3)$	$(4,4)$	$(1,1)$

Since $\Gamma = -1$, the unnormalized backgrounds are

$$\mathcal{B}_{X_3} = \mathcal{B}_{X_4} = 2(u - u^{-1})(v - v^{-1}) = -8\sin\alpha\sin\beta. \tag{C9}$$

The other sixteen backgrounds vanish. All eighteen defect contributions still equal $2(\cos\alpha - \sin\beta)$, so equal signed components are equivalent to $\sin\alpha\sin\beta = 0$ for all real parameters. In a neighborhood of the reference point, these are exactly the two coordinate axes. Changing only the zero-string phase cannot remove their difference: X_2 and X_3 belong to the same class and have identical defect terms.

3. Phases of four input groups and an independent zero-string phase

This section classifies all regular measurements near the reference point when each of the four input groups has an independent phase and the physical zero-string phase can be adjusted separately. All signed components are positive at the reference point. By Appendix B, in a sufficiently small neighborhood the common-radius tetrahedral conditions are therefore equivalent to equality of these components.

For this stronger local statement, allow arbitrary phases on the four values of (q_A, q_B). After removing a global phase, write

$$\lambda_{00} = 1, \quad \lambda_{10} = e^{i\alpha}, \quad \lambda_{01} = e^{i\beta}, \quad \lambda_{11} = e^{i\gamma}. \tag{C10}$$

The seed is

$$\frac{-i}{8} C_6 \left[\sum_z i^{|z|} \lambda_{q_A(z), q_B(z)} |z\rangle + (e^{i\phi} - e^{i\gamma}) |0^6\rangle \right]. \tag{C11}$$

The reference point is $(\alpha, \beta, \gamma, \phi) = (0, 0, 0, \pi/2)$. The direct superposition has $\gamma = \alpha + \beta$ and $\phi = \pi/2$.

The defect contributions before the factor $2/64$ are

$$F_I = \sin(\phi - \alpha) - \sin(\gamma - \alpha),$$
$$F_{II} = \cos(\gamma - \beta) - \cos(\phi - \beta). \tag{C12}$$

The complete unnormalized backgrounds follow from the finite sum

$$\mathcal{B}_R = \Gamma \sum_{z\in\mathbb{F}_2^6} (-1)^{h\cdot z} \lambda_{q_A(z),q_B(z)} \overline{\lambda_{q_A(z\oplus a),q_B(z\oplus a)}}. \quad \text{(C13)}$$

With the pullbacks of Table II and the explicit Boolean functions above, the only nonzero expressions are

$$\begin{aligned} \mathcal{B}_{X_3} = \mathcal{B}_{X_4} &= 4[-\cos\alpha + \cos\gamma \\ &\quad - \cos(\alpha-\beta) + \cos(\beta-\gamma)], \\ \mathcal{B}_{Y_3} &= 8[\cos\alpha - \cos(\beta-\gamma)], \\ \mathcal{B}_{Z_4} = \mathcal{B}_{Z_5} = \mathcal{B}_{X_6} &= -8[\sin\beta + \sin(\alpha-\gamma)]. \end{aligned} \quad \text{(C14)}$$

The other twelve expressions vanish identically. Thus each actual signed expectation is $(\mathcal{B}_R + 2F_I)/64$ or $(\mathcal{B}_R + 2F_{II})/64$, according to its class.

In particular, X_1 and X_2 have zero background and belong to classes I and II, respectively. Equal signed components therefore require $F_I = F_{II}$ and force each expression in Eq. (C14) to vanish. Conversely, these conditions are sufficient. The background equations reduce to

$$\begin{aligned} \sin\beta &= \sin(\gamma-\alpha), \\ \cos\alpha &= \cos(\gamma-\beta), \\ \cos\gamma &= \cos(\alpha-\beta). \end{aligned} \quad \text{(C15)}$$

Near the origin, the first equation gives $\gamma = \alpha+\beta$, because sine is injective near zero. Substitution into the last equation gives

$$\cos(\alpha+\beta) - \cos(\alpha-\beta) = -2\sin\alpha\sin\beta = 0.$$

Hence either $\beta = 0$ and $\gamma = \alpha$, or $\alpha = 0$ and $\gamma = \beta$. Both branches satisfy the middle equation.

Finally, let $G = F_I - F_{II}$. At the reference point, $\partial_\phi G = -1$, and $\phi = \pi/2$ solves $G = 0$ along both branches. The implicit function theorem makes this solution unique nearby. The complete local solution set is therefore

$$(\alpha, 0, \alpha, \pi/2) \quad \text{or} \quad (0, \beta, \beta, \pi/2). \quad \text{(C16)}$$

Equation (C16) therefore classifies the local regular measurements in the stated four-group phase model and proves the local $n = 6$ branch statement in Sec. V.

Appendix D: Low-dimensional proofs

This appendix supplies the rigidity and curved-path proofs for Sec. VI, followed by the invariant that distinguishes the specified four-qubit reference measurement from the known larger-radius segment.

We first fix the phase-expectation formulas used throughout the local calculations. Write $b_z = i^{f_0(z)}$ and let $\theta_0 = 0$. For a signed physical Pauli operator $\widetilde{P}$, define the fixed matrix coefficients

$$K_P(z,w) = \overline{b_w} b_z \langle w|C_n^\dagger \widetilde{P} C_n|z\rangle. \quad \text{(D1)}$$

The seed component and all its derivatives follow from the finite identity

$$2^n m_P(\theta) = \sum_{z,w} K_P(z,w) e^{i(\theta_z - \theta_w)}. \quad \text{(D2)}$$

In particular,

$$2^n \partial_l m_P(0) = i \sum_{z,w} K_P(z,w)(\mathbf{1}_{z=l} - \mathbf{1}_{w=l}). \quad \text{(D3)}$$

These formulas use the original circuit and therefore keep its signs and normalization explicit.

1. Local rigidity at $n = 3$

This section proves local uniqueness of the three-qubit reference measurement by an invertible Jacobian minor.

For $n = 3$, let $\mathcal{F}$ contain the eight differences $m_P - m_{X_1}$, in the order

$$P = Y_1, Z_1, X_2, Y_2, Z_2, X_3, -Y_3, Z_3.$$

The columns correspond to $\theta_1, \ldots, \theta_7$. Equation (D3) gives

$$4D\mathcal{F}(0) = \begin{pmatrix} -2 & -2 & 0 & 1 & 1 & 0 & 2 \\ -1 & -2 & 0 & -1 & 2 & 2 & 0 \\ -1 & -1 & -1 & 0 & 1 & 0 & 1 \\ -1 & -1 & -1 & 0 & 1 & 1 & 0 \\ -1 & -2 & 0 & 1 & 0 & 0 & 2 \\ -2 & -1 & 0 & 1 & 0 & 0 & 2 \\ 0 & -2 & -1 & 1 & 2 & 0 & 0 \\ -1 & 0 & -2 & 1 & 0 & 2 & 0 \end{pmatrix}. \quad \text{(D4)}$$

The determinant of the first seven rows of the displayed integer matrix is 4. Hence the corresponding derivative determinant is $4/4^7 = 1/4096$. The inverse function theorem applies to these seven real analytic component differences and implies that their common zero is locally unique. Every regular-tetrahedral measurement sufficiently near the specified point satisfies these equations: all signed components are positive there, so equality of their squares implies equality of the components themselves. This proves Proposition 1 in the full seven-dimensional phase space.

2. All straight phase paths with real generators at $n = 4$

This section classifies the real directions whose straight phase paths satisfy the tetrahedral constraints, first by necessary derivative conditions and then by substitution into the full equations.

a. Linearization. Let $\mathcal{F}$ now contain the eleven differences from X_1, with the remaining signed operators ordered as

$$Y_1, Z_1, X_2, Y_2, Z_2, X_3, Y_3, Z_3, X_4, -Y_4, Z_4.$$

For a line $\theta = td$, Eq. (D2) gives the derivative rule

$$16\frac{d^k}{dt^k} m_P(td)\Big|_{t=0} = \sum_{z,w} K_P(z,w)\big[i(d_z - d_w)\big]^k. \quad \text{(D5)}$$

The first-order equations have precisely the following four-parameter solution, where a, b_*, c, κ are arbitrary real numbers:

$$\begin{aligned} d_0 = 0, \qquad & d_1 = d_2 = d_9 = a, \\ d_3 = d_8 = a - c, \qquad & d_{10} = b_* - c, \\ d_4 = d_5 = -a + b_* - c/2 + \kappa, \qquad & \\ d_6 = d_7 = d_{12} = d_{13} = c, & \\ d_{11} = b_*, \qquad & d_{14} = d_{15} = \kappa. \end{aligned} \quad \text{(D6)}$$

This follows from a linear-algebra reduction of Eq. (D3): all four displayed directions annihilate $D\mathcal{F}(0)$, and the minor of $16D\mathcal{F}(0)$ with columns $(1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14)$ has determinant 65536. Thus its rank is 11 and the displayed kernel is exhaustive.

b. Higher-order necessary conditions. Put $G_P(t) = 16[m_P(td) - m_{X_1}(td)]$. The second-order conditions from Eq. (D5) include

$$\begin{aligned} G''_{Y_1}(0) &= -2c(2a - c), \\ G''_{Y_2}(0) &= 4c(a - 2b_* + 2c). \end{aligned} \quad \text{(D7)}$$

If $c \neq 0$, their vanishing requires $a = c/2$ and $b_* = 5c/4$. The next condition becomes

$$G''_{X_2}(0) = -\frac{c(c + 4\kappa)}{2} = 0, \quad \text{(D8)}$$

so $\kappa = -c/4$. But the same derivative rule then gives

$$G'''_{Y_1}(0) = -\frac{39c^3}{8} \neq 0. \quad \text{(D9)}$$

Consequently $c = 0$. The remaining necessary second-order condition is

$$G''_{X_2}(0) = -8(a - b_*)(a - b_* - \kappa) = 0. \quad \text{(D10)}$$

c. Sufficiency. Define the three subsets of the integer input labels

$$\begin{aligned} \mathcal{A} &= \{1, 2, 3, 8, 9, 10, 11\}, \\ \mathcal{B} &= \{1, 2, 3, 8, 9, 14, 15\}, \\ \mathcal{C} &= \{4, 5, 14, 15\}. \end{aligned} \quad \text{(D11)}$$

The branch $b_* = a$ gives $d = a\mathbf{1}_{\mathcal{A}} + \kappa\mathbf{1}_{\mathcal{C}}$, whereas the branch $\kappa = a - b_*$ gives $d = b_*\mathbf{1}_{\mathcal{A}} + (a - b_*)\mathbf{1}_{\mathcal{B}}$. Substitution into the finite sum (D2) gives, for every signed local Pauli operator and either branch,

$$16m_P(td) = e^{iat} + e^{-iat}. \quad \text{(D12)}$$

This verifies the full constraints for all real parameters and completes the sufficiency proof. Hence all allowable directions are exactly

$$\mathrm{span}_{\mathbb{R}}\{\mathbf{1}_{\mathcal{A}}, \mathbf{1}_{\mathcal{C}}\} \cup \mathrm{span}_{\mathbb{R}}\{\mathbf{1}_{\mathcal{A}}, \mathbf{1}_{\mathcal{B}}\}. \quad \text{(D13)}$$

Local regularity implies the signed equations used above by positivity at the reference point. Their differences along any real direction are finite exponential sums, so equality on an open interval extends to the whole line. Equations (D12) and (D13) establish the first part of Theorem 4 without a restriction on the number of values taken by d.

There are fifteen phase variables after removal of the global phase, and the eleven component differences have Jacobian rank eleven at the reference point. The implicit function theorem therefore makes their local zero set a four-dimensional smooth manifold. Positivity of the signed components identifies it with the nearby common-radius tetrahedral measurements. The directions that extend to admissible straight lines form only the two planes in Eq. (D13); the next section constructs a nonlinear phase path.

3. The curved branch and its full monotonic interval

This section verifies three properties: the chosen phases reduce the local constraints to $H = T$; this equation has a unique target root branch through the origin on the stated interval; and the common radius increases strictly along that branch.

a. Reduction to the scalar constraint. Equation (D14) specifies all phases in Eq. (42). Its entries are indexed by $z = 0, \ldots, 15$, with the most significant bit first. Inserting $\theta_z = bu_z + \delta v_z$ into Eq. (D2) gives precisely the two expressions in Eq. (40). In particular, these identities hold for independent real b, δ, before any branch is selected.

z	0	1	2	3	4	5	6	7
u_z	0	1	1	1	2	2	0	0
v_z	0	1	1	0	1	1	1	1
z	8	9	10	11	12	13	14	15
u_z	1	1	0	0	0	0	1	1
v_z	0	1	0	1	1	1	0	0

(D14)

b. Existence and identification of the root branch. Set $\eta = -\delta$, $y = \tan(\eta/2)$, and $x = \tan(b/2)$. The half-angle identities give

$$H - T = \frac{2(Ay^2 - By + C)}{1 + y^2}. \quad \text{(D15)}$$

We now verify the root in Eq. (41), including its entire domain. Put $k = \sqrt{2} - 1$. Then

$$Ay^2 - By + C = \frac{x+1}{(1+x^2)^2}\mathcal{P}(x,y), \tag{D16}$$

where

$$\begin{aligned}\mathcal{P}(x,y) = &-x^3y^2 + x^3y - 3x^2y^2 - 3x^2y - 4x^2 \\ &+ 3xy^2 + xy + 4x + y^2 - 3y.\end{aligned} \tag{D17}$$

To control the root as b varies, first estimate the derivative in x. On the square $0 \le x, y \le k$, one has

$$\mathcal{P}_{xx} = 6y[x(1-y) - (1+y)] - 8 < 0. \tag{D18}$$

Using $k^2 + 2k - 1 = 0$ and $4 - 12k < 0$, it follows that

$$\begin{aligned}\mathcal{P}_x(x,y) &\ge \mathcal{P}_x(k,y) = 4 - 8k + (4-12k)y \\ &\ge 20k - 8 = 20\sqrt{2} - 28 > 0.\end{aligned} \tag{D19}$$

Uniqueness of the root follows from strict decrease in y. The other derivative satisfies

$$\begin{aligned}\mathcal{P}_y &= 2y(1 + 3x - 3x^2 - x^3) + x^3 - 3x^2 + x - 3 \\ &\le 2k(1+3k) + k^3 + k - 3 = 5 - 4\sqrt{2} < 0.\end{aligned} \tag{D20}$$

Existence and endpoint values follow from the boundary signs:

$$\begin{aligned}&\mathcal{P}(x,x) = -x(1+x^2)(x^2+2x-1) \ge 0, \\ &\mathcal{P}(x,k) \le \mathcal{P}(k,k) = 0.\end{aligned} \tag{D21}$$

Thus each $x \in [0,k]$ has a unique root $y(x) \in [x,k]$; $y(0) = 0$, $y(k) = k$, and $x < y(x) < k$ in the interior. At the root, $\mathcal{P}_y < 0$. Since $A > 0$, the corresponding quadratic has positive discriminant and this is its smaller root. Rationalizing that root gives

$$y(x) = \frac{2C}{B + \sqrt{B^2 - 4AC}}, \tag{D22}$$

which is well defined also at $x = 0$. This identifies the root with the stated closed formula throughout the interval.

c. Strict increase of the common radius. The derivative signs above also determine the motion of the simple root. Differentiation gives $y'(x) = -\mathcal{P}_x/\mathcal{P}_y > 0$. Consequently $\eta'(b) > 0$ and $b \le \eta(b) \le \pi/4$. With $T = \cos(b-\eta) + \sin\eta$,

$$T'(b) = \sin(\eta - b) + \eta'(b)\,[\cos\eta - \sin(\eta - b)] > 0. \tag{D23}$$

The bracket is strictly positive: equality would require both $\eta = \pi/4$ and $b = 0$, inconsistent with the determined endpoint values. The first term is nonnegative. Finally, $T(0) = 1$ and $T(\pi/4) = 1 + 1/\sqrt{2}$ give the image (39). At the origin, $y'(0) = 4/3$, $\eta'(0) = 4/3$, and $r_4'(0) = \sqrt{3}/6$.

Setting $n = 4$ and $m = T/8$ in Eq. (10) gives all reduced states, with each fixed tetrahedral direction occurring four times. The unitary factorization in Eq. (42) gives a complete orthonormal measurement, and $b = 0$ recovers the specified D_{f_0}.

4. Non-equivalence of the reference measurement to the known $n = 4$ segment

We show that the specified reference measurement is not equivalent, under local unitaries, qubit permutations, or outcome relabeling, to any measurement on the $n = 4$ segment in Appendix C of Ref. [13]. Equivalent measurements must have the same one-qubit spectra and the same sum of the three $2|2$ purities. For a pure state at $n = 4$ define

$$S(\psi) = \mathrm{Tr}(\rho_{12}^2) + \mathrm{Tr}(\rho_{13}^2) + \mathrm{Tr}(\rho_{14}^2). \tag{D24}$$

The sum is invariant under local unitaries and qubit permutations: complementary subsystems of a pure state have identical nonzero spectra, so a permutation only permutes the three unordered $2|2$ terms. Each column of a Pauli-orbit measurement shares these invariants with its seed. Outcome relabeling therefore cannot change the comparison.

To test these necessary conditions, we write the earlier family explicitly. With $a \in [0, \pi/3]$, let $x = e^{ia}$ and write its input phases as $i^{f_z} x^{k_z}$. In order $z = 0, \ldots, 15$, the two fixed vectors are

$$\begin{aligned}f &= (0,0,0,3,0,0,1,0, \\ &\qquad 0,0,0,2,0,1,3,3), \\ k &= (0,1,-1,-1,0,1,0,0, \\ &\qquad -1,2,0,1,3,3,1,0).\end{aligned} \tag{D25}$$

Here $a = \pi(1-\tau)/3$ in the parameterization of that reference. The seed $C_4(i^{f_z}x^{k_z})_z/4$ has one-qubit eigenvalues $(1 \pm \sqrt{3}\,c(x))/2$, where

$$c(x) = \frac{x^3 + x^{-3} + 2x + 2x^{-1}}{16} = \frac{\cos 3a + 2\cos a}{8}. \tag{D26}$$

This is nonnegative on the stated segment, decreases from $3/8$ to zero, and gives the already known endpoint $3r_E(4)$.

At the specified f_0 point the three purities are $(63, 39, 39)/128$, and hence $S_0 = 141/128$.

On this segment, equality of the one-qubit spectrum and of S requires $c = 1/8$ and $S = 141/128$, respectively. Evaluating the reduced-state purities from the amplitudes in Eq. (D25) and clearing denominators, with $x \ne 0$, gives the necessary equations $P(x) = Q(x) = 0$, with

$$\begin{aligned}P(x) &= x^6 + 2x^4 - 2x^3 + 2x^2 + 1, \\ Q(x) &= 6x^{16} - 3x^{14} + 74x^{12} + 48x^{10} \\ &\quad + 134x^8 + 48x^6 + 74x^4 - 3x^2 + 6.\end{aligned} \tag{D27}$$

The Euclidean algorithm excludes a common root. Its first remainder is

$$\begin{aligned}Q \bmod P = &\,816x^5 - 2340x^4 + 2168x^3 \\ &- 1240x^2 + 616x - 596,\end{aligned} \tag{D28}$$

whose gcd with P is 1. Thus the original parameter segment contains no measurement equivalent to the specified f_0 measurement under the allowed operations.

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